

# Environmental costs of a fully isolated AI.

Posted by u/Suspicious-Host9042 • 0 points • 12 comments

Consider this, a laptop running a locally-run AI model, not connected to the internet, charged using solar power (they do exist), not connected to the power grid, just generating images on its own without any outside resources.

Is this possible? If so, how much water and electricity does it use?

You might argue about the costs of making the laptop and solar panels, and I agree about that. But just like how artists don't really take into account the costs of making the paint, paper, etc. that they use to draw on, I don't think we should consider the costs of making the tools either.

## Comments

u/seraphinth • 1 points • 2025-10-16 05:13 UTC

downvoted but not commented by anti-s because this is out of their bot script for their arguments

u/Pretend\_Jacket1629 • 2 points • 2025-10-16 04:14 UTC

it's pure and simple virtue signaling over a fabricated "issue"

image generation is hundreds if not thousands of times less energy than equivalent tasks made without it

in fact, basically no appliance in your house uses less than tens if not hundreds of times more by merely being plugged in each day

[https://www.reddit.com/media?](https://www.reddit.com/media?url=https%3A%2F%2Fpreview.redd.it%2Ffw0bjg6r9ruf1.png%3Fwidth%3D1387%26format%3Dpng%26auto%3Dwebp%26s%3D0a50a018aed9daa61061228)

[url=https%3A%2F%2Fpreview.redd.it%2Ffw0bjg6r9ruf1.png%3Fwidth%3D1387%26format%3Dpng%26auto%3Dwebp%26s%3D0a50a018aed9daa61061228](https://www.reddit.com/media?url=https%3A%2F%2Fpreview.redd.it%2Ffw0bjg6r9ruf1.png%3Fwidth%3D1387%26format%3Dpng%26auto%3Dwebp%26s%3D0a50a018aed9daa61061228)

it's such a microscopic amount of energy that people are getting harassed over, and yet none of this is equally towards any other activity- even when the alternative to ai is almost always going to take more energy

u/Purple\_Food\_9262 • 2 points • 2025-10-16 03:52 UTC

It's kind of interesting, as in your case clearly the solar infrastructure would be totally sustainable. However, despite what people like you claim around here that local gen is better, there's a pretty legit point that data centers are more environmentally friendly and efficient than computers connected to the grid. It's very often repeated here that local gen is more efficient though.

Giant microwave section breaks it then with actual numbers about efficiency vs vibes <https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>

u/Normal-Room5279 • 0 points • 2025-10-17 13:44 UTC

To be perfectly honest, I find it hard to believe.

Simply due to the fact that AI data centers need to be operating 24.7

So every subscriber will be able to access it at any time

While a machine runs locally, only needed to be accessed by a single person.

Which also means you can turn it off at some point.

I also would imagine that it would require more intense cooling.

Simply because having more computers Producing that much heat in a single area will have a compounding effect.

u/Purple\_Food\_9262 • 1 points • 2025-10-17 15:40 UTC

That's fine if you don't believe it, I don't really have time to cover the information you would need to know to understand it, but that's what the studies say and the math checks out. Have a good one.

u/Purple\_Food\_9262 • 1 points • 2025-10-16 03:52 UTC

It's kind of interesting, as in your case clearly the solar infrastructure would be totally sustainable. However, despite what people like you claim around here that local gen is better, there's a pretty legit point that data centers are more environmentally friendly and efficient than computers connected to the grid. It's very often repeated here that local gen is more efficient though.

Giant microwave section breaks it then with actual numbers about efficiency vs vibes <https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>

u/perusing\_jackal • 2 points • 2025-10-16 03:30 UTC

I think at that stage the arguing about energy consumption would just be silly. Opensource and local run models are better for the environment and if people truly care about this issue, stop using online subscription based sites and use a local model. You don't need

anything past 20b to spell check, do an internet search or reword some text. Most tasks can be done by GPT-oss just fine.

u/Thick-Protection-458 • 2 points • 2025-10-16 12:26 UTC

\> Opensource and local run models are better for the environment

Worse, more probably.

Underutilized hardware still consume some power (not so relevant in this version) + need to be produced.

u/perusing\_jackal • 0 points • 2025-10-16 13:14 UTC

Why are people trying to argue about if local running model are more efficient. Either you have 1 machine on for 3 hours a day running local ai when the user wants to ask it a question and then it's tuned off the rest of the day. Alternatively you have users computer on plus a huge datacentre on 24/7 with all the infrastructure required to route all the internet traffic just waiting for a user to make a request.

How on gods green earth, are people really trying to tell me the latter is more energy efficient?

u/Purple\_Food\_9262 • 1 points • 2025-10-17 16:18 UTC

Do you have any studies that back up that local inference has higher efficiency than data centers? The chip efficiency of the data center hardware alone is already 2-3x that of consumer hardware, and basically no one is running batch inference at home either.

u/perusing\_jackal • 0 points • 2025-10-17 16:33 UTC

ffs, how can a data centre being on 24/7 be more efficient than not using any electricity most of the day, at all, because you have your computer unplugged? you really need a study to work that out? sometimes it feels like people here agree just for the sport of it.

u/Thick-Protection-458 • 2 points • 2025-10-17 17:14 UTC

\> not using any electricity most of the day, at all

Okay, now lets do some freakin mental math than.

Assuming  $\backslash \text{UserCount} \backslash$  people sending  $\backslash \text{QueryCount} \backslash$  queries at any given second (which is largely true, while not exactly).

So you are a cloud provider with  $\backslash \text{ActiveProviderMachines} \backslash$  active machines over time, and these C machines activated in such an amount so utilization are close to 100%.

Assuming even energy efficiency of the compute being the same (not true, compute-oriented GPU have kinda higher, as far as I am aware).

Now assume each (average) query processing needs  $\backslash \text{PowerRequiredPerQuery} \backslash$  watts.

Now assuming average user machine needs  $\backslash \text{PowerRequiredForUserMachine} \backslash$  watts.

Lets compute how much energy is required to process all these requests now.

$$E_{\text{providergpu}} = \text{QueryCount} \backslash * \text{PowerRequiredPerQuery}$$

$$E_{\text{usermachines}} = \text{QueryCount} \backslash * \text{PowerRequiredForUserMachine}$$

$$E_{\text{providertotal}} = E_{\text{providergpu}} + E_{\text{usermachines}} = \text{QueryCount} \backslash * \text{PowerRequiredPerQuery} + \text{QueryCount} \backslash * \text{PowerRequiredForUserMachine}$$

Now assuming, due to that same underutilization, generating the same response for user query required some higher amount of power. Like

$$\text{PowerRequiredPerNonbatchedQuery} = \text{PowerRequiredPerQuery} \backslash * k, \text{ where } k \text{ is like... well, for instance } 1.05$$

Only thing that changes than...

$$E_{\text{localtotal}} = E_{\text{providergpu}} \backslash * k + E_{\text{usermachines}} = \text{QueryCount} \backslash * \text{PowerRequiredPerQuery} \backslash * k + \text{QueryCount} \backslash * \text{PowerRequiredForUserMachine}$$

Where, due to  $k \geq 1.0 \rightarrow E_{\text{localtotal}} > E_{\text{providertotal}}$

\-----

\> not using any electricity most of the day, at all

Oh, and one more thing...

Your own machine can be turned off, sure.

Provider machines serve another user queries in the same time.  
So in a manner of speaking they're offloading that (k-1.0) \\* PowerRequiredPerQuery additional energy demand from someone all the time.

And...

Providers do not necessary have to keep all their machines turned on all the time too. You can turn them on or off depends on demand.

u/Purple\_Food\_9262 • 2 points • 2025-10-17 16:35 UTC

Ok, I mean for you this is a vibes based thing I guess. I mean in simple terms it's like you're claiming public transportation is less efficient than everyone having their own car. I thought those of us on the pro side would be more based in facts but it takes all kinds I guess. Have a good one.

u/perusing\_jackal • 0 points • 2025-10-17 16:59 UTC

ok, figure me this, I am saying you use an llm locally like once or twice a day. that means you computer is on for 5 mins. The rest of the time you have no computer on. no datacenter, no electricity draw. I get your trying to play some batch processing of multiple user mind gymnastics but I can only control my actions in the world not what other people do. If I use local gen, there is a 100% guarantee of less energy being used, because my computer is turned off most of the day. I don't need a scientific paper to figure out 0 is less than any positive number.

u/Thick-Protection-458 • 2 points • 2025-10-17 17:20 UTC

\> rest of the time you have no computer on. no datacenter

The rest of the time datacenter process other user queries

u/Purple\_Food\_9262 • 1 points • 2025-10-17 17:00 UTC

Not interested in bro science, have a good one.

u/Thick-Protection-458 • 2 points • 2025-10-16 13:33 UTC

Because first one will have more time slots of underutilization when hardware would still consume some power.

Look, the two facts

- that during single query processing you are memory bandwidth constrained rather than compute constrained
- that is you use vllm and batch processing - you get much more throughput per batch (not per query)

By itself means there during single query processing are some ALUs in your GPU which do not compute anything now. But nevertheless they need some power, although less than actively processing stuff.

If you have a batch workload you can surely minimize that. But usually you don't have such workload.

So energy required to process N queries as standalone queries individually is more than what required to process same N queries in batch.

**u/Normal-Room5279** • 0 points • 2025-10-17 13:48 UTC

Not sure I completely understood everything is sad. But I didn't understand the bottom line.

That according to your data centers are more efficient in fulfilling user queries. (At least I think it's what you meant. Correct me from wrong.)

But even so won't the sheer amount of queries plus the huge size of the data center would completely overshadow that kind of efficiency.

**u/Thick-Protection-458** • 2 points • 2025-10-17 14:53 UTC

> That according to your data centers are more efficient in fulfilling user queries

Or at least that, from energy perspective - it is *\\*expected\\** to do same computations in batched mode in datacenters than in single-query mode on user machines. But essentially - yes.

> But even so won't the sheer amount of queries

Well, the way I understood the first comment guy was like "same workload done on user's own machines", means same amount of queries - just X queries coming from Y users / pipelines processed on same Y machines or on (simplifyingly) one giant machine.

> plus the huge size of the data center

May add additional challenges, but that's kinda complicated.

**u/DaylightDarkle** • 2 points • 2025-10-16 03:22 UTC

> Is this possible

Yes.

If you want local image generation, invoke is probably the simplest to set up for that.

Anyhow, anyone correct me if I'm wrong, most of the resource cost comes from the training, while using it afterwards is relatively cheap.

**u/perusing\_jackal** • 2 points • 2025-10-16 03:37 UTC

It's the datacenters that are the issue. GPT, Gemini and others have large server racks that need cooling. Companies have started building datacenters by rivers to feed directly in to the cooling. Obviously this does not apply just to AI but people are concerned that as AI is one of the most visited sites on the internet, it's just adding to the problem. Local AI would reduce a large portion of the problem, but people like convenience so won't take the time to learn how to use it and just complain about the datacenters being bad.

**u/Purple\_Food\_9262** • 1 point • 2025-10-16 03:51 UTC

It's kind of interesting, as in your case clearly the solar infrastructure would be totally sustainable. However, despite what people like you claim around here that local gen is better, there's a pretty legit point that data centers are more environmentally friendly and efficient than computers connected to the grid. It's very often repeated here that local gen is more efficient though.

Giant microwave section breaks it then with actual numbers about efficiency vs vibes  
<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>